The Comparability of Airbnb Review Signals

An Integrated Analysis of London Reviews, Listing Quality and Expectation Mismatch

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Abstract

Airbnb ratings are often treated as transparent measures of listing quality, yet review scores are produced through linguistic conventions, guest expectations and platform-specific reviewing norms. Using **2,097,795 London Airbnb reviews**, this report examines how listing-score structure, review language, sentiment, aspect emphasis and negative-review content jointly shape the interpretation of guest evaluations.

The analysis combines language detection, VADER sentiment scoring, multilingual BERT sentiment modelling, keyword-based aspect tagging, non-parametric statistical tests and a rule-based plus TF–IDF logistic regression classifier for negative reviews. The central argument is that Airbnb reviews combine property quality with reviewing style: the language of the comment, the aspects guests mention and the type of complaint all condition what a score can reasonably be taken to mean.

English accounts for most reviews, but the 17.9% non-English share is large enough to change how the corpus should be read. Korean-reviewed listings have the highest average overall score ($M = 4.773$), while Spanish-reviewed listings have the lowest ($M = 4.679$). Same-listing VADER comparisons also score English reviews more positively, although the size of those gaps points as much to model sensitivity as to guest behaviour. Aspect tagging adds a different layer: communication and location vary most across language groups, while value varies least. In negative reviews, classifiable complaints more often describe deficiencies than mismatches, but many cases remain unresolved.

1 Introduction

Airbnb reviews transform private accommodation experiences into public evaluative signals. Prospective guests encounter ratings, textual comments and host information as evidence of trustworthiness, while hosts and platforms depend on the same signals for reputation and ranking. This comparability is built into the platform format, but it cannot be taken for granted analytically.

A four-star review may signal disappointment in one reviewing culture and satisfaction in another. Reviewers also differ in how much they explain: one guest documents a maintenance failure, another leaves a single word of praise and another criticises a stay that matched the listing but not their expectations. These differences matter because Airbnb rankings, host reputations and guest decisions depend on signals that look more objective than they are.

The analysis begins with the structure of the London Airbnb market: compressed ratings, class imbalance, room-type differences, borough patterns and temporal change. It then turns to the language of reviews, asking whether ratings, sentiment and aspect emphasis vary across linguistic groups. The final layer examines negative reviews, where the key distinction is between a concrete listing deficiency and an expectation mismatch.

The guiding question is:

> How should Airbnb reviews be interpreted when scores, text and complaints are shaped by both listing quality and cultural-linguistic reviewing patterns?

Four research questions follow from this:

1. Are listing-level scores associated with the language in which reviews are written?
2. Do sentiment differences remain when different language groups review the same listing?
3. Which aspects of the stay do different language groups emphasise?
4. Among negative reviews, are complaints more often linked to observable deficiencies or to expectation mismatch?

Two constraints shape the interpretation from the start. Airbnb does not report reviewer nationality and Inside Airbnb provides listing-level scores rather than individual review ratings. Language is used as a proxy for cultural-linguistic background, not as a direct measure of nationality or culture.

2 Framing and Related Work

Online reviews are both measurements and messages. Prior work treats review systems as reputation mechanisms that reduce information asymmetry, but also shows that ratings are shaped by platform incentives, selection effects and consumer expectations [1, 2, 3]. The numerical score compresses the stay into a platform-readable signal, while the written review reveals what the guest noticed, expected and chose to emphasise.

Airbnb makes this tension especially visible because listings differ widely in room type, location, host professionalism and price. Expectation-disconfirmation theory is useful here: dissatisfaction can arise either because the service is poor or because the experienced stay falls short of what the guest expected [4]. Cross-cultural research also warns that rating behaviour and communication norms are not uniform across groups [5]. In a multilingual corpus, those human differences interact with technical measurement problems, since sentiment models and language detectors do not perform identically across languages [7, 8, 9].

Three tensions structure the empirical design. Ratings are concentrated near five stars, so small differences can matter inside a narrow effective range. Language affects both expression and model interpretation, making it a source of information as well as measurement error. Negative text also has multiple meanings: a complaint about a broken shower points to a different problem than a complaint about a neighbourhood that was accurately described but poorly suited to the guest.

3 Data and Preprocessing

The empirical analysis is based on the public Inside Airbnb London release, combining review text, listing metadata and borough-level geographic boundaries [6]. Because the report uses different subsets for descriptive, linguistic, sentiment and negative-review analyses, Table 1 distinguishes the raw corpus from the analytical samples.

Table 1: Data audit across the integrated project

Dataset stage	Count
Raw review rows in the London reviews file	2,097,996
Final review rows with usable comments after cleaning	2,097,795
Unique reviewers in the review corpus	1,768,660
Listings retained in the main review-score analysis	72,749
Listing metadata records in the September 2025 scrape	96,871
Features in the listing metadata after merge	87 columns
Negative reviews used in deficiency–mismatch analysis	4,943
Stratified sample used for multilingual BERT inference	50,000

The full cleaned review corpus spans December 2009 to September 2025. Some exploratory summaries begin later because they require listing-score fields or annual-count filters; the 2025 figures are partial-year values through September.

The listings file contains host location, room type, price, neighbourhood, availability variables, host variables and seven review-score dimensions: overall rating, accuracy, cleanliness, check-in, communication, location and value. The geographic file provides boundaries for London's 33 boroughs.

Several preprocessing choices are important for interpretation:

- Rows with missing review comments were removed before text analysis.
- Price and percentage variables were parsed into numeric fields for exploratory modelling.
- Review language was detected from comment text.
- Listing-level scores were merged onto reviews, but they are not individual per-review scores.
- Negative reviews were identified using VADER compound sentiment below -0.05.

The listing-level score issue is the main unit-of-analysis constraint. When the report discusses rating differences by language, it means that listings reviewed in those languages have different listing-level score profiles. It does not mean that each individual reviewer assigned the listed score.

4 Dataset Context

4.1 Compressed Rating Distributions

The rating distribution is central to the modelling problem because Airbnb scores are concentrated near the top of the scale, as shown in Figure 1. The mean overall rating is 4.75 out of 5.00, with a standard deviation of 0.23. The median is 4.81, the first quartile is 4.66 and the third quartile is 4.91. This leaves little visible separation between good and excellent listings.

Using a threshold of 4.0 to define low-rated listings gives 69,634 positive listings (≥ 4.0) and 3,115 low-rated listings (< 4.0). This is a 95.7% versus 4.3% split, or an imbalance ratio of 22.4:1. The threshold is strict in the Airbnb context because a rating below four is already unusual. It is therefore useful for isolating weak listings, but it should not be treated as the only possible definition of dissatisfaction.

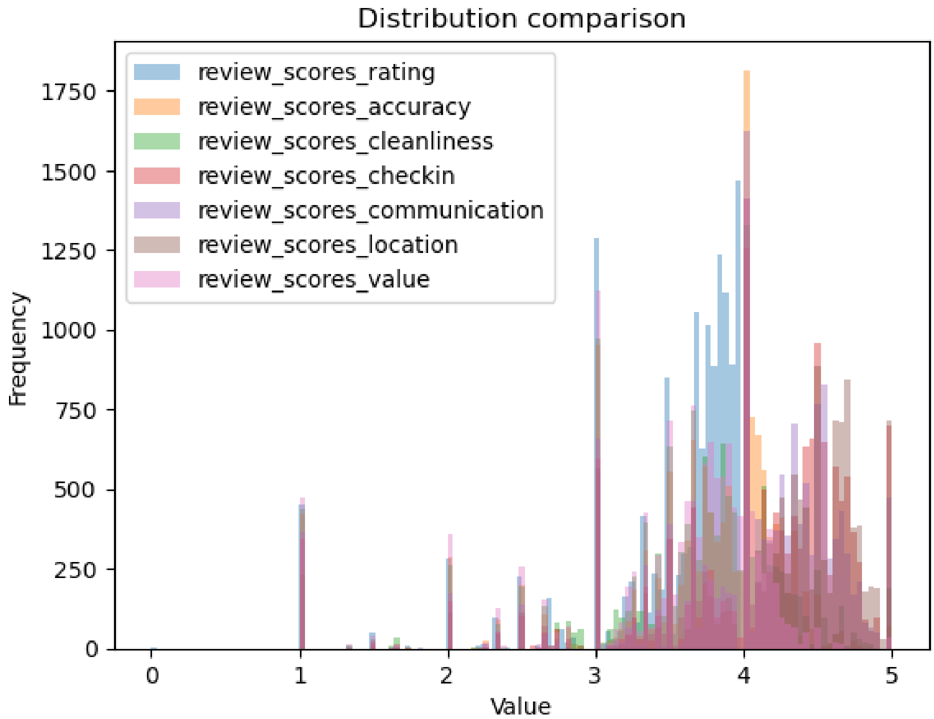

Figure 1: Review-score distributions are concentrated near the upper end of the scale, limiting the discriminatory power of raw ratings.

4.2 Sub-score Discriminability

The seven Airbnb sub-scores do not distinguish weak and strong listings equally. Table 2 shows that value and accuracy have the largest gaps between low-rated and top-rated listings, while location has the smallest. This matters because location is partly known before booking; guests may discount it if the listing description was accurate. Value and accuracy, by contrast, directly measure whether the experience felt worth the price and matched the advertised offer.

Table 2: Mean sub-score differences between low-rated and top-rated listings

Sub-score	Low mean	Top mean	Δ Top – Low
Location	3.79	4.85	1.07
Check-in	3.66	4.93	1.27
Communication	3.55	4.96	1.40
Cleanliness	3.19	4.87	1.68
Accuracy	3.18	4.93	1.75
Value	2.98	4.84	1.86

4.3 Room Type, Borough and Time

Room type is associated with both rating and risk of low scores. Table 3 summarises these differences: shared rooms have the highest low-rated share, while entire homes and private rooms dominate the dataset by volume and have similar average ratings. Hotel rooms are expensive but underperform relative to private listings, although this interpretation should be treated as a hypothesis because guest expectations may differ across accommodation types. Figure 2 visualises the same contrast.

Table 3: Quality and pricing patterns by room type

Room	N	Mean	Low %	Price
Shared room	167	4.24	18.6	93.20
Hotel room	93	4.47	7.5	442.39
Entire home	48,904	4.68	4.3	225.72
Private room	23,585	4.69	4.1	117.81

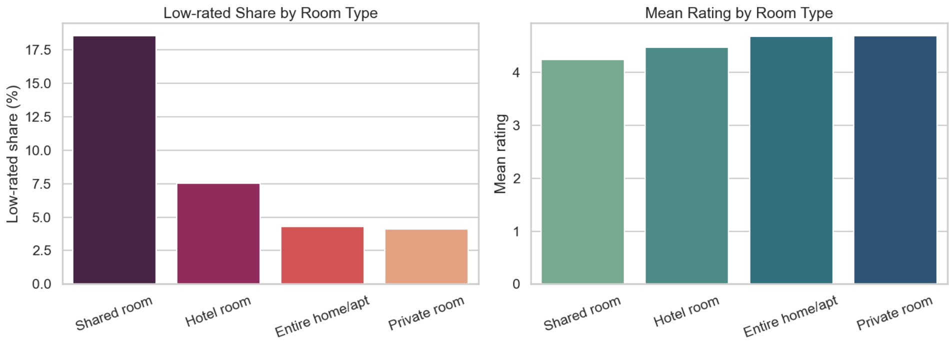

Figure 2: Shared rooms have a much higher low-rated share than entire homes or private rooms.

At borough level, outer London areas such as Redbridge, Barking and Dagenham, Havering, Croydon and Hillingdon have higher low-rated shares in the exploratory analysis, as Figure 3 shows. This may reflect transport access, guest expectations or listing mix, but the available data do not isolate those mechanisms.

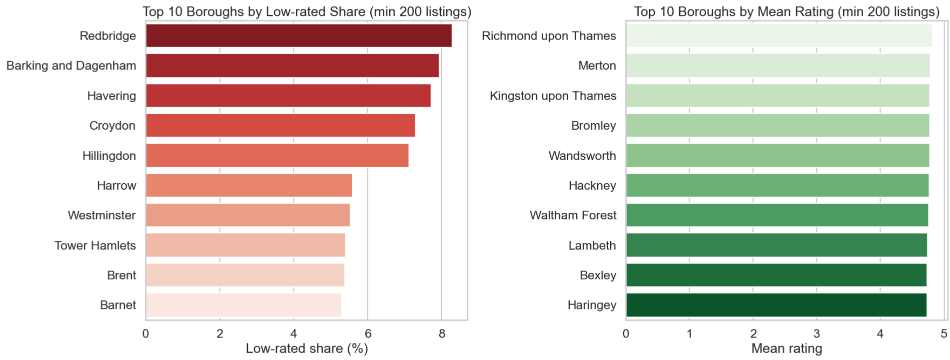

Figure 3: Low-rated listing shares and average ratings vary by borough; the pattern should be interpreted alongside room type and location mix.

The time series also has a clear structural break, plotted in Figure 4. Review volume rises before 2020, falls sharply during the pandemic and recovers strongly from 2022 onward. The 2024 total reaches 450,473 reviews, while the 2025 count of 406,129 covers only the period up to September.

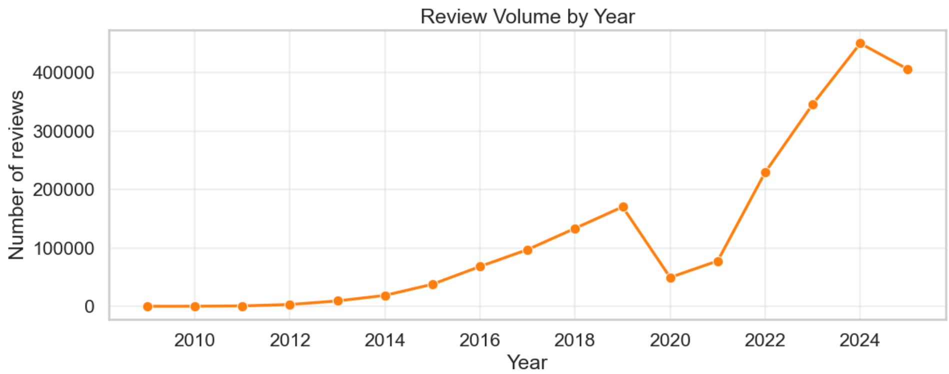

Figure 4: Review volume drops in 2020 and then recovers strongly, which raises the possibility of temporal drift in guest mix and review norms.

The implication is methodological as much as descriptive. When ratings are compressed and metadata carries limited direct signal, the written review becomes the main evidence for how guests interpret the stay.

5 Methods

5.1 Language Detection

Since Airbnb does not provide reviewer nationality, comment language is used as a proxy for cultural-linguistic background. We use `langid`, an off-the-shelf language identification tool [8], restricted to 26 target languages expected in London reviews. Restricting the candidate set improves speed and reduces improbable classifications at this scale.

The proxy has limits. A Spanish-language review may come from Spain or Latin America and a bilingual traveller may choose English for convenience. For that reason, the report refers mainly to language groups rather than national groups.

5.2 Sentiment Models

VADER provides a scalable full-corpus sentiment measure for short review texts [7]. It assigns each review a compound score from -1 to $+1$ and reviews below -0.05 are treated as negative for the deficiency–mismatch analysis.

Its English-oriented lexicon makes it unsuitable as the sole basis for cross-language inference. The nlptown multilingual BERT sentiment model, based on the BERT architecture [9, 10], is therefore used as a robustness check on a stratified 50,000-review sample rather than as a replacement for the full-corpus analysis.

5.3 Aspect Tagging

Reviews are tagged for six aspects aligned with Airbnb score dimensions: cleanliness, location, communication, check-in, value and accuracy. Table 4 lists the aspect categories and example multilingual keywords used for this step.

Table 4: Aspect categories and example keywords

Aspect	Example keywords
Cleanliness	clean, dirty, spotless, hygienic, propre, sauber, limpio
Location	location, area, neighbourhood, station, tube, transport, emplacement, Lage, ubicacion
Communication	host, communication, responsive, helpful, accueil, Gastgeber, anfitrion
Check-in	check-in, key, arrival, lockbox, self check, entrada
Value	value, price, expensive, worth, affordable, prix, Preis, precio
Accuracy	accurate, description, photo, expected, as described, exacto

Keyword tagging gives a scalable count of aspect mentions, not a manual semantic reading. It will miss synonyms, negation and language-specific phrasing, so the results are best used to compare broad emphasis across groups.

5.4 Statistical Testing

Because Airbnb ratings are non-normal and compressed near five stars, the rating analysis uses non-parametric tests. Kruskal-Wallis tests compare distributions across language groups [11]. Dunn's post-hoc tests with Bonferroni correction identify pairwise differences. Wilcoxon signed-rank tests compare paired same-listing sentiment differences [12]. Chi-square tests evaluate whether aspect mention frequency is associated with comment language. Cohen's d and Cramer's V provide effect-size context.

With more than two million reviews, statistical significance is easy to obtain. The discussion therefore emphasises direction, effect size, practical scale and methodological caveats rather than p-values alone.

5.5 Deficiency–Mismatch Classifier

The negative-review analysis asks whether dissatisfaction reflects a likely listing deficiency or an expectation mismatch. Aspect dictionaries identify signals such as cleanliness, maintenance, host service, location fit, value fit, space fit and preference fit. Listing-level context is added through average scores, host variables and recurrence of similar complaints for the same listing.

The rule-based classifier assigns each negative review one of four labels:

- **Deficiency:** quality problems such as maintenance, cleanliness or host-service failures, especially when repeated or aligned with lower listing scores.
- **Mismatch:** fit-related problems, such as location, space or preference issues, for otherwise well-rated listings.
- **Mixed:** evidence of both deficiency and mismatch.
- **Unclear:** insufficient evidence for a confident label.

A TF–IDF logistic regression model is then trained on the rule-labelled deficiency and mismatch cases and used to relabel some unclear reviews only when prediction confidence is high. This model expands heuristic labels; it does not create ground truth. Its value is exploratory and diagnostic.

6 Results by Research Question

6.1 Corpus Composition

Language detection identifies 26 languages. English accounts for 82.13% of the corpus, which means that nearly one in five reviews is non-English. French is the largest non-English language group at 5.55%, followed by Spanish and German. Table 5 gives the top languages and Figure 5 visualises the same imbalance.

Table 5: Top 10 detected review languages

Language	Region label	N	Share (%)
English	Anglophone	1,722,987	82.13
French	Western Europe	116,389	5.55
Spanish	Latin/Iberian	67,264	3.21
German	Western Europe	55,725	2.66
Italian	Western Europe	28,495	1.36
Chinese	East Asia	23,725	1.13
Korean	East Asia	16,375	0.78
Portuguese	Latin/Iberian	16,260	0.78
Dutch	Western Europe	14,321	0.68
Japanese	East Asia	5,892	0.28

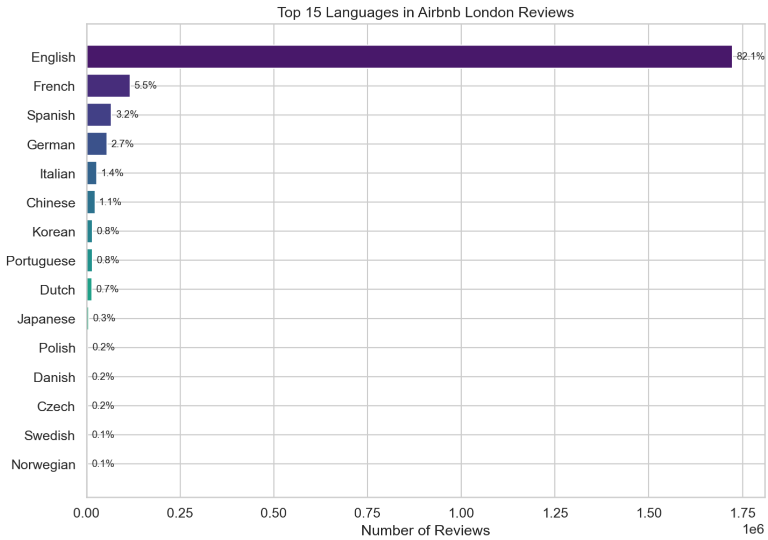

Figure 5: English dominates the review corpus, but the non-English share is large enough to analyse cross-language patterns.

6.2 RQ1: Listing-Level Scores by Review Language

Listings reviewed in different languages have different listing-level score profiles. Korean-reviewed listings have the highest average overall score ($M = 4.773$), followed by English ($M = 4.762$) and Chinese ($M = 4.755$). Spanish-reviewed listings have the lowest average overall score ($M = 4.679$), followed by Portuguese ($M = 4.699$) and French ($M = 4.705$). Table 6 reports the Kruskal-Wallis results across all seven rating dimensions.

Table 6: Kruskal-Wallis tests across language groups

Rating dimension	H-statistic	p-value
Overall rating	21,295	$< .001$
Accuracy	19,609	$< .001$
Cleanliness	19,682	$< .001$
Check-in	10,668	$< .001$
Communication	11,347	$< .001$
Location	13,941	$< .001$
Value	18,549	$< .001$

Dunn's post-hoc test identifies 39 significant pairwise differences after Bonferroni correction. The strongest contrasts are English–Spanish, English–French and Chinese–Spanish. These are listing aggregates, so the pairwise differences shown in Figure 6 should not be read as individual reviewer scores.

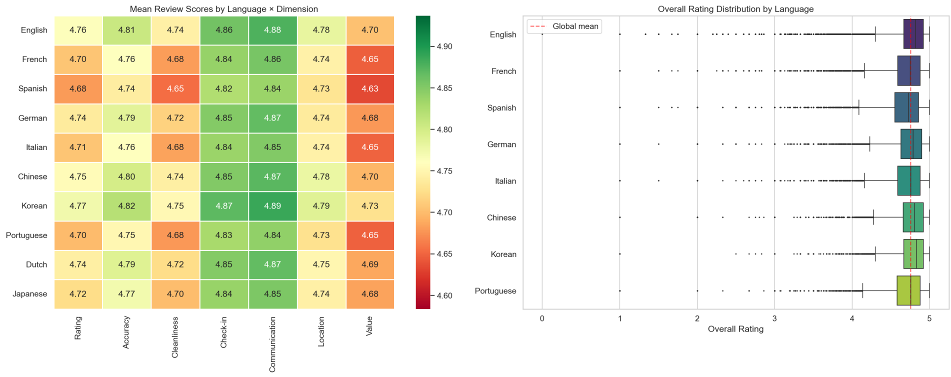

Figure 6: Mean listing scores vary by review language within Airbnb's compressed high-rating range.

6.3 RQ2: Same-Listing Sentiment

Score differences by language could come from booking patterns rather than review expression. Guests writing in different languages may choose different neighbourhoods, room types, price bands or seasons. The same-listing comparison narrows this problem by holding the property constant: it uses listings with at least three English reviews and at least three reviews in another target language, then compares average sentiment for the same property. Table 7 gives the paired VADER comparisons.

Table 7: Same-listing paired VADER sentiment comparison

Lang.	**Listings**	Δ	d	p
French	11,992	+0.651	3.78	$< .001$
German	6,252	+1.254	4.08	$< .001$
Spanish	7,045	+0.769	4.06	$< .001$
Italian	2,867	+0.617	3.46	$< .001$
Dutch	1,115	+0.726	3.30	$< .001$
Portuguese	1,357	+0.711	3.87	$< .001$

English reviews are more positive than non-English reviews for the same listings across all six comparisons. Figure 7 shows that the direction is consistent, but the magnitude warrants caution: Cohen's d values above 3 are more plausibly interpreted as a combination of expression differences, model bias and scale effects than as direct evidence of satisfaction differences.

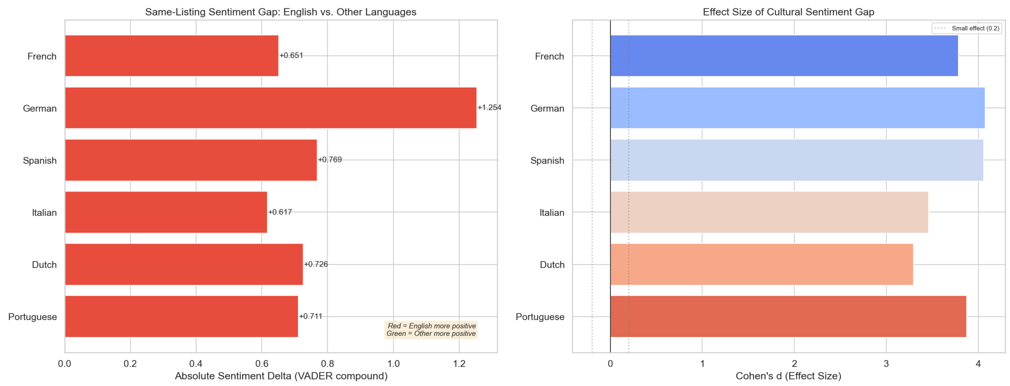

Figure 7: Same-listing VADER comparisons produce English-positive gaps across all six language comparisons.

6.4 Multilingual BERT Check

The multilingual BERT model reduces dependence on VADER's English lexicon by predicting star sentiment on a stratified 50,000-review sample. Figure 8 shows that its distributions still differ by language, supporting cross-language variation without reproducing VADER's exact effect sizes.

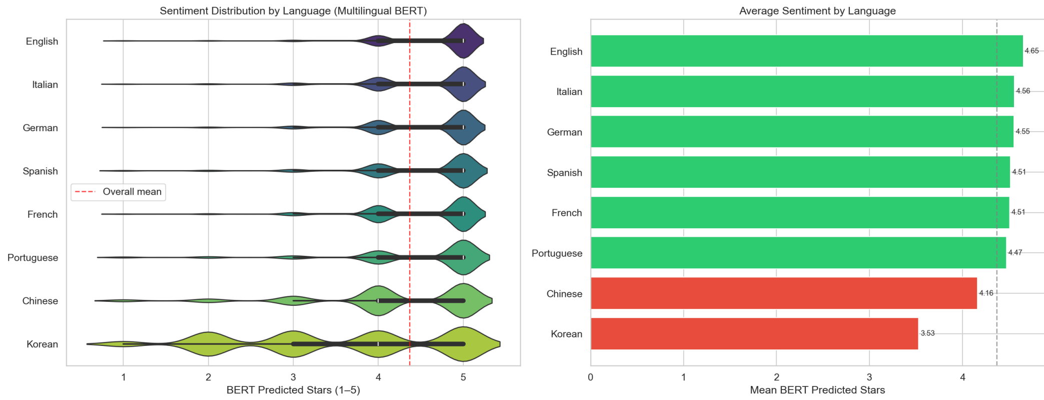

Figure 8: Multilingual BERT sentiment distributions provide a model-based check on cross-language variation.

Multilingual transformers still have weaknesses on short reviews, mixed-language comments, sarcasm and politeness formulas. A stronger validation design would combine multilingual BERT, translated-to-English sentiment scoring and a small human-coded benchmark.

6.5 RQ3: Aspect Emphasis by Language

Language groups differ not only in positivity, but also in what they discuss. Table 8 shows significant associations between review language and all six aspect categories. Communication and location have the largest effects, while value has the weakest.

Table 8: Aspect mention frequency by language: chi-square tests

Aspect	χ^2	Cramer's V	Interpretation
Communication	162,140	0.280	Medium effect
Location	115,508	0.236	Small–medium
Check-in	73,311	0.188	Small–medium
Cleanliness	35,878	0.132	Small effect
Accuracy	24,397	0.109	Small effect
Value	5,778	0.053	Negligible

The aspect results in Figure 9 explain part of the language gap. Communication and location are experience categories where expectations and description styles can diverge, whereas value appears to be a more shared concern across groups.

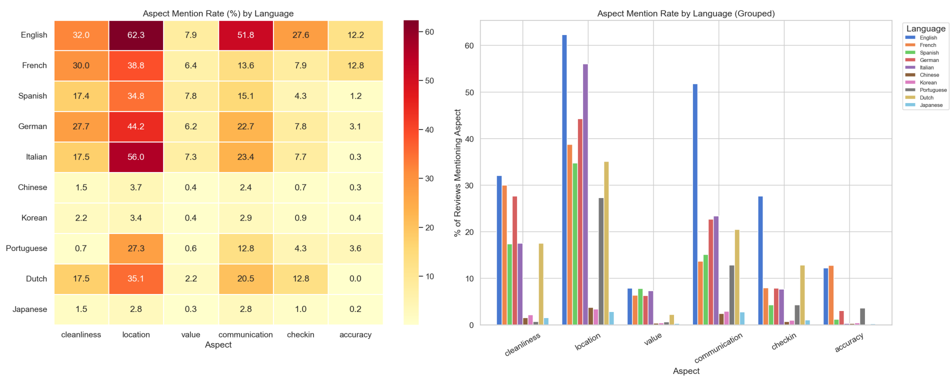

Figure 9: Communication and location carry the strongest language-associated variation in aspect mentions.

6.6 RQ4: Negative Reviews as Deficiency or Mismatch

The deficiency–mismatch analysis examines 4,943 negative reviews. Table 9 shows that the rule-based classifier is intentionally conservative: 77.65% of negative reviews are initially unclear. After the TF–IDF logistic regression expansion, the unclear share falls to 66.92%. Deficiency rises from 10.32% to 20.96%, while mismatch remains almost unchanged at 9.29%.

Table 9: Negative-review label shares before and after text-model expansion

Label	Rule-based (%)	Expanded (%)
Unclear	77.65	66.92
Deficiency	10.32	20.96
Mismatch	9.20	9.29
Mixed	2.83	2.83

The logistic regression model was trained on 965 labelled negative reviews and reached 0.8031 holdout accuracy. It then evaluated 3,838 unclear reviews and reassigned 530 of them under the confidence threshold. Because the model learns from rule-derived labels, the expansion shown in Figure 10 refines the heuristic system rather than validating it against independent ground truth. Its main contribution is narrow: it reallocates more cases to deficiency without producing a comparable rise in mismatch.

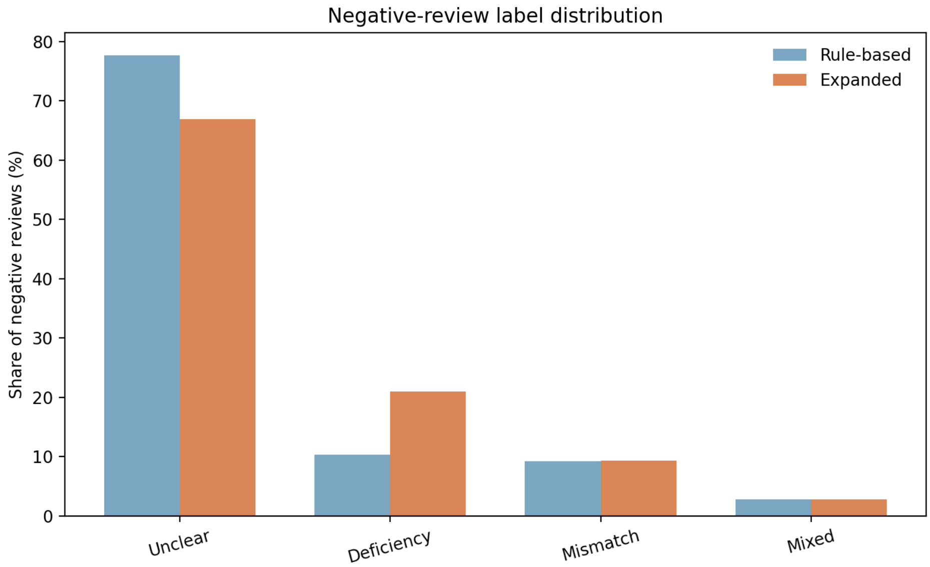

Figure 10: Expanded labels reduce the unclear category primarily through additional deficiency classifications.

Aspect frequencies support the interpretation of the labels, as Figure 11 shows. Deficiency reviews are most associated with host service problems (26.74%), maintenance (22.39%) and cleanliness (11.68%). Mismatch reviews are dominated by location fit (80.83%), with smaller roles for space fit (25.71%) and preference fit (21.13%). Mixed reviews contain both types of signals.

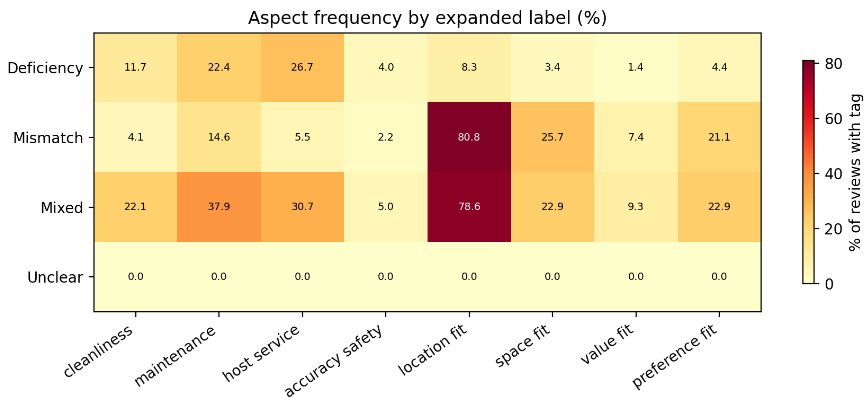

Figure 11: Aspect frequencies separate quality-related complaints from fit-related complaints.

Average listing ratings also align with the distinction. Figure 12 shows that reviews labelled mismatch come from listings with an average rating of 4.74, close to unclear reviews at 4.75. Deficiency reviews are attached to lower-rated listings on average (4.68) and mixed reviews are lower still (4.61). The pattern is consistent with the intended distinction, but it is not independent validation because listing scores also inform the labelling rules.

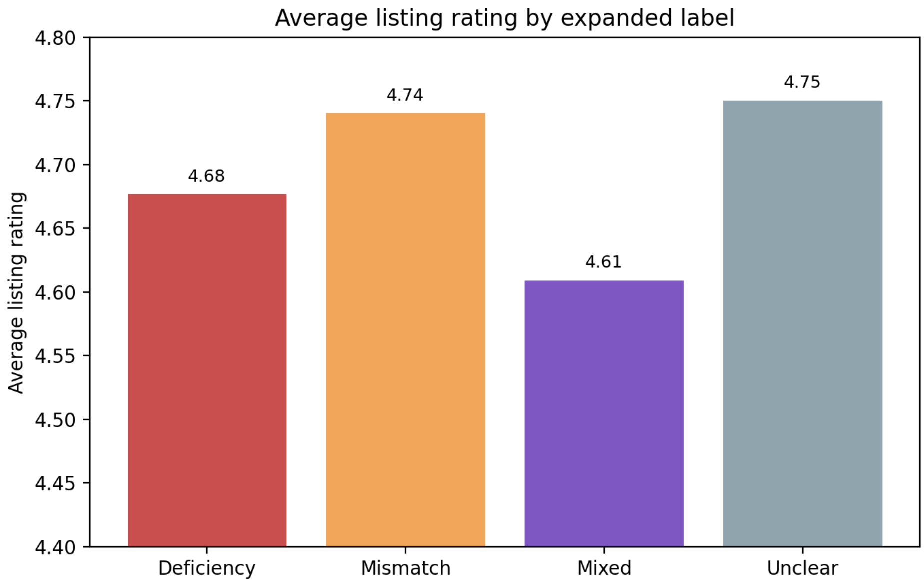

Figure 12: Average listing ratings are lower for deficiency and mixed labels than for mismatch and unclear labels.

6.7 Supplementary Context: Geography, Host Country and Review Length

The review-language pattern is geographically uneven. Central boroughs such as Southwark, Camden and Westminster attract the highest non-English review shares, while outer boroughs such as Sutton, Havering and Bromley are more English-dominated. Figure 13 maps this pattern and French is the leading non-English review language in every London borough.

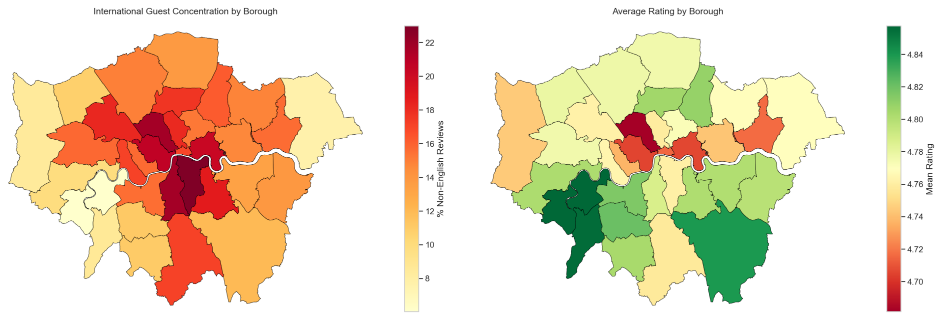

Figure 13: Non-English review share is highest in central London boroughs, while outer boroughs are more English-dominated.

Host-country analysis is less conclusive. As Figure 14 shows, UK-based hosts dominate the London market, accounting for 74.6% of hosts with usable location fields. Some guest–host cultural clustering is visible, but the pattern is mostly shaped by the large number of UK hosts. The evidence is therefore insufficient to claim a strong cultural-affinity mechanism.

Figure 14: UK-based hosts dominate the market; host-country differences are secondary to the overall structure of the London supply.

Review length changes how much textual material each subcorpus provides, as shown in Figure 15. German-language reviews are longest, with a median of 36 words. French and English are also relatively detailed, with medians of 34 and 32 words. Chinese and Japanese reviews have a median of one word and Korean reviews fall between these extremes at 23 words. Cross-language comparison should account not only for sentiment polarity, but also for differences in elaboration.

Table 10: Mechanisms that may affect Airbnb review comparability

Mechanism	**How it affects reviews**	**Main implication**
Listing quality	Properties differ in cleanliness, accuracy, maintenance, value and host service.	Some score gaps may reflect real differences in the stays being reviewed.
Guest selection	Language groups may book different boroughs, room types, price bands or seasons.	Same-listing comparison reduces, but does not remove, selection bias.
Expectation formation	Guests compare the stay with what they expected from price, location and description.	Negative text can describe mismatch rather than objective failure.
Review expression	Reviewer groups differ in directness, length, praise and criticism.	Star scores and sentiment are partly communicative conventions.
Model measurement	Sentiment and language tools perform unevenly across languages and short texts.	Cross-language NLP results need validation before behavioural interpretation.

Table 11: Robustness and validation status

Issue	**Current treatment**	**Stronger validation needed**
Listing-level scores	Caveat stated and rating language softened.	Re-run rating tests at listing-language level or use clustered/mixed-effects models.
Selection effects	Same-listing sentiment comparison.	Add controls or slices by borough, room type, price band, season and post-2020 period.
VADER language bias	Interpreted as diagnostic rather than definitive sentiment evidence.	Repeat same-listing comparison with multilingual BERT or translated reviews.
BERT sentiment	Stratified 50,000-review sample and distribution figure.	Add per-language means, confidence intervals and paired same-listing tests.
Language detection	Restricted `langid` classifier.	Manual audit of a stratified sample, especially short and mixed-language reviews.
Aspect tagging	Multilingual keyword dictionaries.	Hand-code a validation set and report precision, recall and false positives.
Deficiency–mismatch labels	Conservative rule system plus TF–IDF expansion.	Validate against blind manual annotation and report class-level precision/recall/F1.

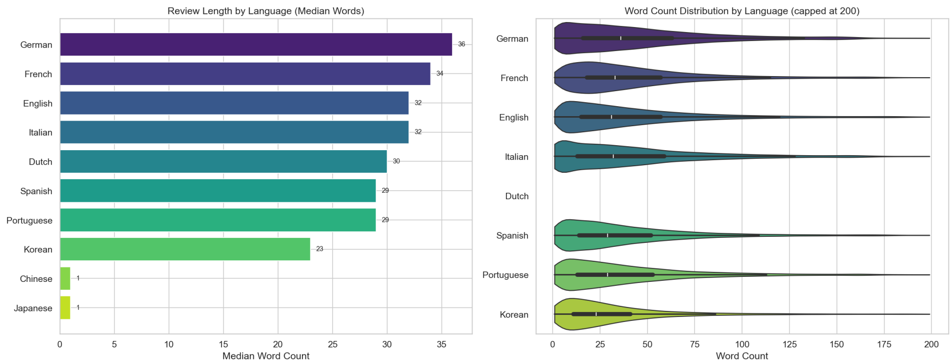

Figure 15: Review length varies sharply by language, affecting how much textual evidence is available for sentiment and aspect analysis.

7 Mechanisms and Robustness

The empirical results point to associations, not a single causal pathway. Table 10 separates the main mechanisms that can produce language-associated review differences. This distinction is important because the same observed gap can arise from property quality, guest selection, expectation formation, expression style or measurement bias.

Several reviewer agents highlighted that a full 20/20 version would need additional robustness checks beyond the outputs available in the three submitted reports. Rather than presenting uncomputed checks as completed results, Table 11 records which issues are addressed by the current report and which remain as necessary validation work.

These checks do not undermine the descriptive findings, but they determine how far the claims can be pushed. The strongest conclusions are about rating compression, language-associated textual differences and the need for contextual interpretation. Claims about cultural causality, true satisfaction gaps or the exact prevalence of mismatch remain exploratory.

8 Discussion

The central issue is comparability. Airbnb presents ratings as if they can be averaged across guests, listings and languages, yet several transformations occur before a stay becomes a score. Ratings are compressed near five stars; review text varies in length and aspect focus; sentiment models react differently across languages; and negative reviews mix repairable deficiencies with expectation failures.

These findings do not imply that Airbnb reviews are unreliable; rather, they show that their evidentiary value is conditional. A lower average score for one language group may reflect stricter expression, different booking patterns, model bias or real differences in the listings selected by that group. A negative comment about location may be less a sign of poor quality than of mismatched expectations. A complaint about maintenance or host service, especially when repeated for the same listing, carries a different operational meaning.

The platform implication is therefore specific: review summaries should separate three questions that are currently blended together. What happened at the property? Which part of the stay did the guest emphasise? Is the complaint recurring across guests, or does it reflect a fit problem for a particular guest profile? A host dashboard built around these distinctions would be more useful than a single average rating and a guest-facing summary could explain whether criticism concerns cleanliness, communication, location fit or value.

Rating normalisation by reviewer language is more controversial. It may reduce systematic differences in reviewing style, but it may also hide legitimate dissatisfaction or treat direct criticism as a bias to be corrected. The stronger recommendation is contextualisation: show rating distributions, aspect-specific summaries and language-sensitive sentiment evidence without automatically discounting any group's evaluations.

9 Limitations

Several limitations are already built into the interpretation, but they remain material for any extension of the work.

- **Language proxy.** Review language does not identify culture, nationality or residence.
- **Rating aggregation.** Listing-level scores merged onto reviews can overstate significance if treated as independent review-level ratings.
- **Model dependence.** VADER is English-centric, multilingual BERT is still imperfect and keyword aspect tags need manual validation.
- **Operational labels.** The deficiency–mismatch classifier expands rule-derived labels rather than verified ground truth, leaving many negative reviews unresolved.
- **Remaining selection.** Same-listing comparison controls for property identity, but not guest type, season, trip purpose, room availability, price paid or prior expectation.
- **Temporal drift.** Review norms before the pandemic may differ from post-pandemic norms and 2025 is only a partial year.

10 Conclusion

The London Airbnb corpus shows how much information is lost when reviews are reduced to average stars. Korean-reviewed listings sit at the top of the observed score range and Spanish-reviewed listings at the bottom, but that pattern sits inside a compressed rating scale and relies on listing-level scores. English reviews are more positive in same-listing VADER comparisons, yet the size of the gap reveals as much about language-sensitive sentiment modelling as it does about guest satisfaction. Communication and location vary most in aspect mentions, suggesting that groups differ in how they frame the stay, not only in how positive they sound.

The negative-review analysis adds the final distinction. Among reviews that can be classified, deficiencies are more common than mismatches, but the unresolved share is large. That is a useful finding precisely because it resists a simple conclusion: many complaints cannot be confidently reduced to either bad quality or unrealistic expectations.

The central implication is that Airbnb reviews should be read as contextual evidence rather than as uniform measurements. Review language, textual detail, aspect emphasis and complaint type all condition what an average rating can reasonably be taken to mean. Future work should manually validate aspect labels, repeat the same-listing tests with multilingual BERT and compare London with other cities.

A Reproducibility Notes

The analyses were implemented in Python 3.11 using pandas, matplotlib, seaborn, scipy, scikit-posthocs, GeoPandas, Folium, VADER, langid, transformers and PyTorch. The source files correspond to the Inside Airbnb London review file, listing metadata file and borough GeoJSON boundary file from the September 2025 release. Multilingual BERT inference used an NVIDIA GeForce RTX 3050 Laptop GPU with 4 GB VRAM, CUDA 12.4 via PyTorch 2.6 and FP16 precision. The full workflow, including data loading, language detection, VADER sentiment, BERT inference, aspect tagging, statistical testing and plot generation, completed in approximately 43 minutes on the project laptop.

B Supplementary Figures

The appendix keeps the supporting diagnostics that are useful for audit but secondary to the main argument: the metadata correlation structure in Figure 16, language trends in Figure 17, guest–host affinity in Figure 18, borough language mix in Figure 19, seasonal language patterns in Figure 20, positive and negative word clouds in Figure 21 and raw versus filtered token counts in Figures 22 and 23.

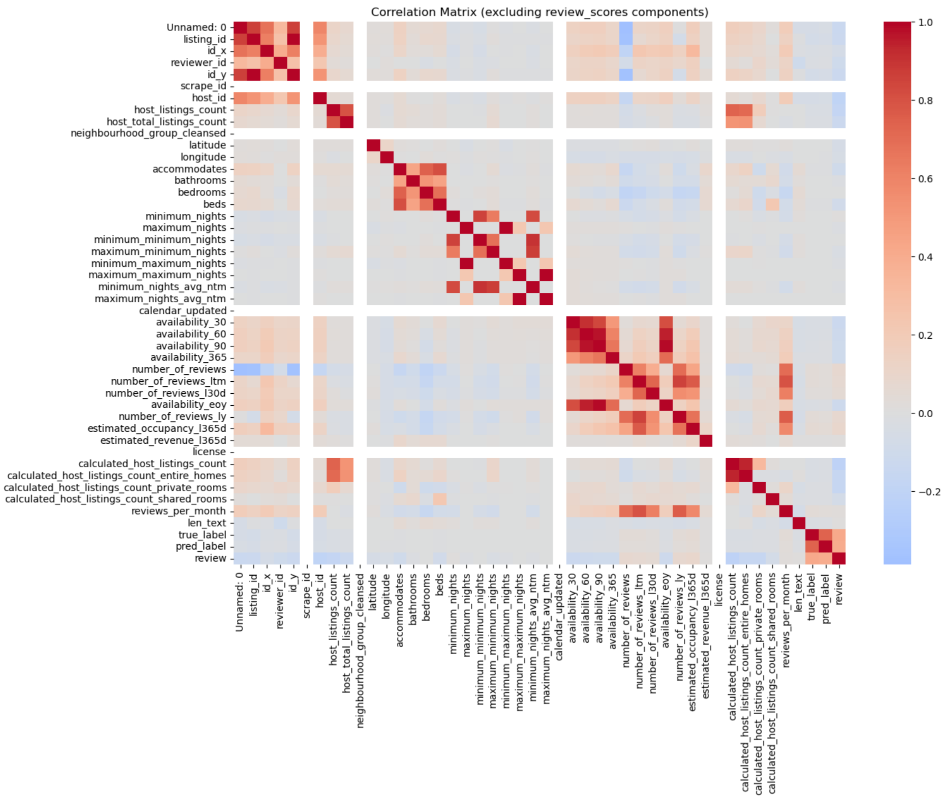

Figure 16: Correlation matrix excluding review-score components. Numerical metadata clusters internally but has limited direct correlation with rating labels.

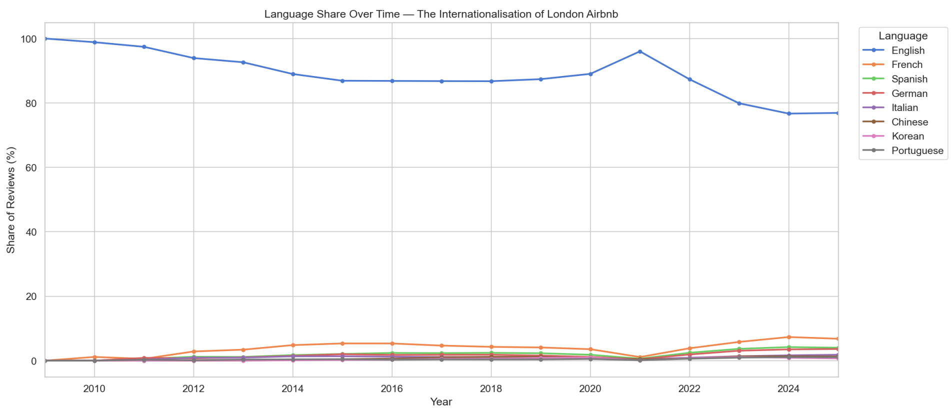

Figure 17: Review-language shares change over time, with non-English reviews becoming more visible after 2015.

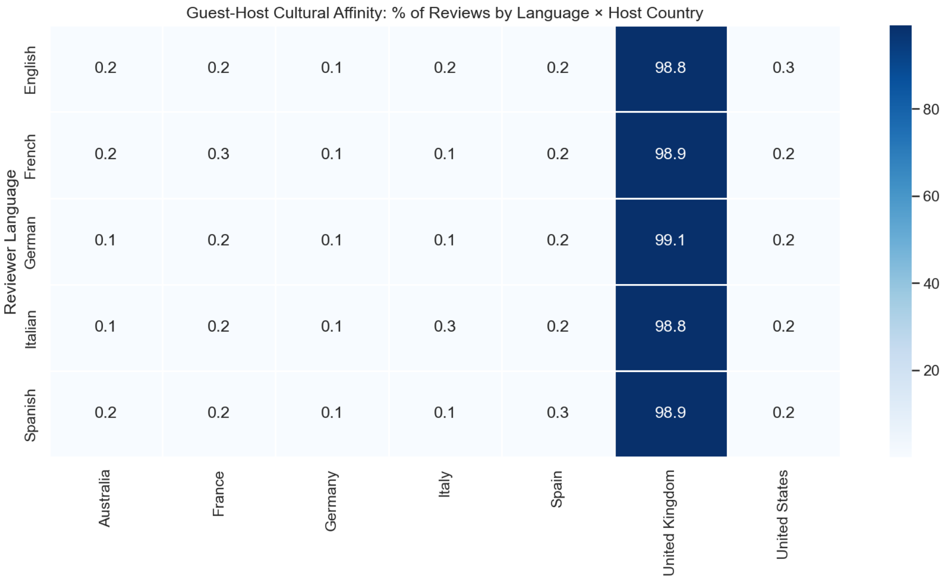

Figure 18: Guest–host cultural affinity matrix. The apparent clustering is dominated by the large UK-host base.

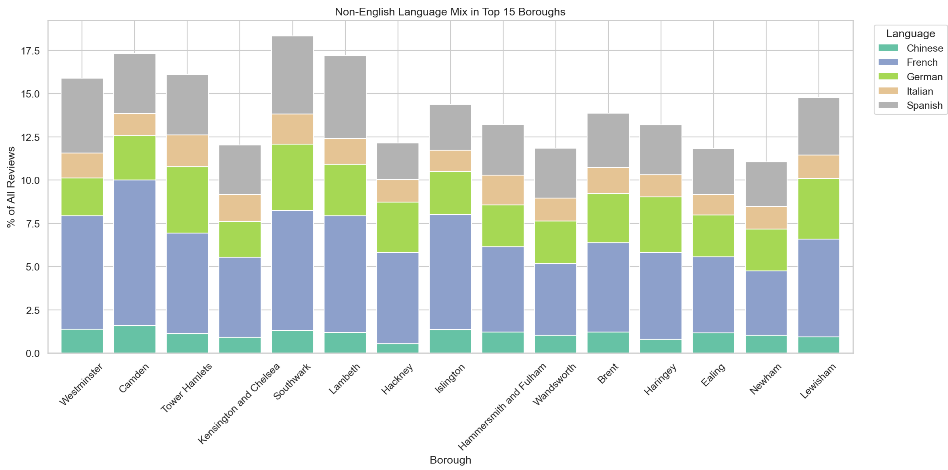

Figure 19: Non-English language composition in high-volume boroughs. French is the leading non-English language across boroughs.

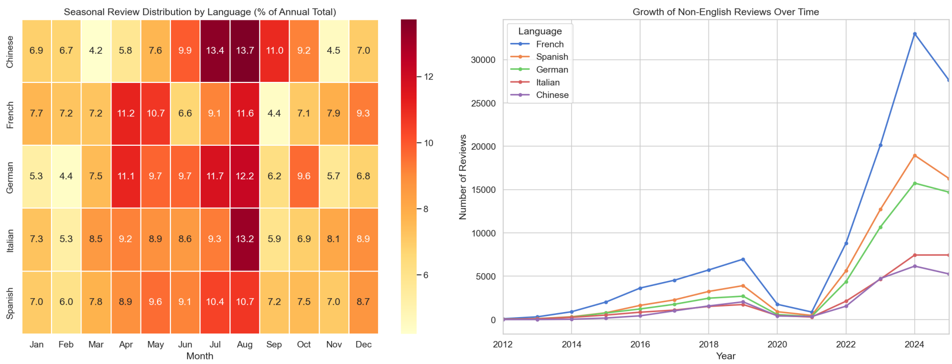

Figure 20: Seasonal patterns and long-term growth of non-English reviews.

Figure 21: Positive and negative word clouds are exploratory diagnostics only; they are sensitive to language mix and stop-word choice.

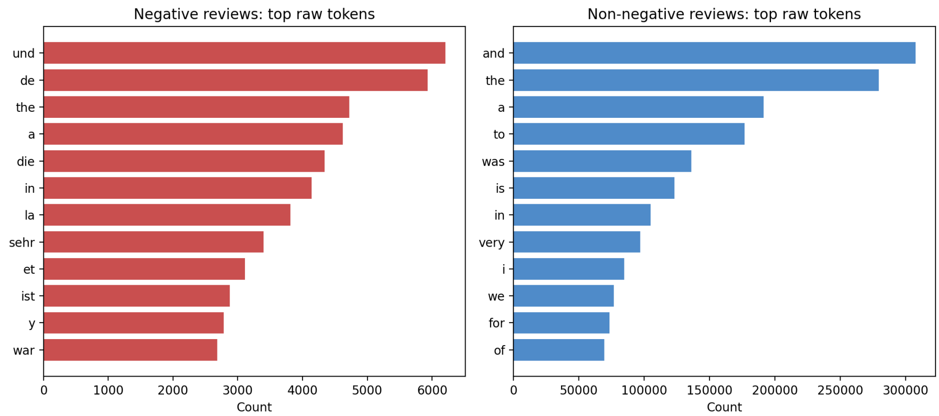

Figure 22: Raw token counts reveal the multilingual composition of the negative-review subset, motivating language-aware filtering.

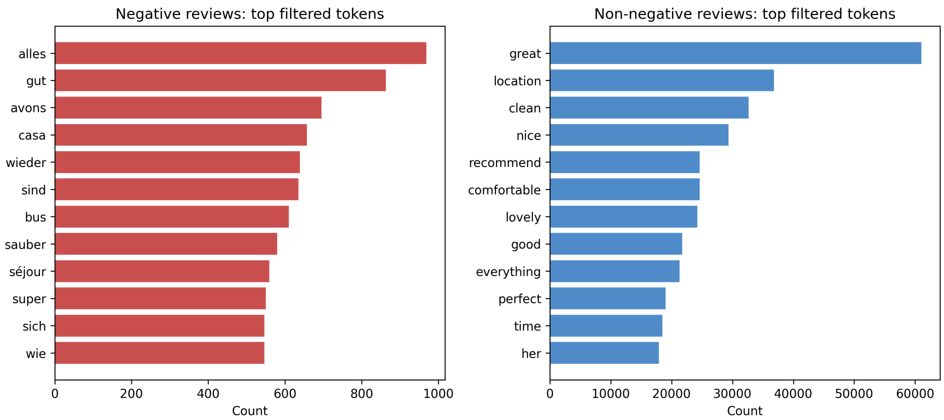

Figure 23: Filtered token counts are more interpretable than raw counts, but still remain descriptive rather than confirmatory evidence.

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